

Visionary

WEIGHT 2 · BEHAVIOR + follow-ups · NO Discord/Drive

One-liner

UMI turns “in-the-wild” from robot logistics into human logistics — the missing portable action-rich demo pipe that later VLAs still cite when they need wrist-aligned, latency-honest manipulation data.

Talk arc

BEHAVIOR prior → poles → cites → follow-ups → apps

1 BEHAVIOR data-engine prior Household LH stress needs cheap scene diversity

2 Complementary poles UMI || DROID || ALOHA

3 $\pi_{0.5}$ / GROOT cite CLEAR from local fulltexts

4 Follow-ups Embodiment-aware · FM · RTC · mobile · MP4

5 New apps Homes/restaurants · cross-robot libraries · dynamic

6 Close Human logistics one-liner · exam bite

BEHAVIOR — data-engine prior

Handheld scaling when you cannot ship Frankas

BEHAVIOR-class benchmarks stress wash / fold / multi-step kitchen — UMI already shows 70% dish & cloth.

Relative wrist-centric EE + DP/FM chunks align with π_0 / $\pi_{0.5}$ / OpenPI control habits.

For 2026 BEHAVIOR: UMI-style handheld scaling = cheap scene diversity without robot logistics.

Complementary pole to DROID robot-in-wild and ALOHA lab teleop — pick the collection axis you can afford.

Do NOT ship 2024 ResNet DP as the 2026 entry — race on $\pi_{0.5}$ / N1.7 with UMI-class demos.

Complementary poles

UMI · DROID · ALOHA — who enters the wild?

UMI

Human logistics

Handheld gripper
anywhere humans go
action-rich demos

DROID

Robot logistics

Franka fleet in
buildings · 76k/564
scene entropy

ALOHA

Lab teleop

Leader-follower
joint-space
fine bimanual

$\pi_{0.5}$ / GROOT CLEAR-cite UMI

Later VLAs still need UMI-class wrist demos

$\pi_{0.5}$ 2504.16054
Bibliography cites UMI — CLEAR local fulltext

GROOT-N1 2503.14734
Cites UMI — CLEAR local fulltext

Deploy forks umi-on-legs / umi-arx
Community proof of interface portability

TENTATIVE π_0 / OpenVLA / Octo
Often cite DP/OXE/DROID more; UMI = interface citation

Follow-ups (1/2)

Research directions that inherit the interface

1. Embodiment-aware policies

Consume kinematically-invalid demos instead of post-hoc drop (paper lim. #1)

2. Textureless-room recovery

Third-person + gripper fiducials when SLAM dies (paper #2)

3. Lighter / higher-DoF handheld

Dex hand while keeping observation alignment

4. DP → flow matching swap

π_0 -style FM under same UMI interface; KEEP latency matching

Follow-ups (2/2)

RTC · mobile · distributed MP4

5. RTC / soft-inpaint async chunks

On top of UMI's latency model for high-Hz arms

6. Mobile manipulator + relative EE

Paper already moves base mid-rollout (Fig. 10 moral)

7. Internet-scale distributed MP4

Paper §IX vision + SLAM QA filters for quality

8. BEHAVIOR diversity card

Steal UMI collection protocol + ablation discipline onto modern heads

New applications

Rapid skill cloning
Where human logistics wins

Homes / restaurants without
robot install during teach

Cross-robot libraries

One demo set → UR / Franka / ARX

Dynamic sports-like

Tossing / sorting beyond
quasi-static BC

Bimanual deformable

Fold + articulated appliances as
VLA FT stress tests

Campus / cafe ZS

Cheap scene cards before Franka
relocation

Education / course

Reproducible handheld BC recipe
under \$400

Motion for BEHAVIOR / VLA era

Keep relative + latency; upgrade the head

Ta=6

UMI DP default

Relative SE(3)

keep across FM swap

Latency match

first-class forever

$\pi_{0.5}$ / N1.7

race on modern heads

ANTI-PATTERN

Shipping 2024 ResNet Diffusion Policy as the 2026 BEHAVIOR entry — steal UMI's INTERFACE and race modern FM/VLA heads.

Fig. 9 — existence proof for cheap wild diversity

1400 demos / 30 locs / 12 person-hours → 71.7% ZS



Human logistics scales scene entropy without Franka fleets

Narrative spine

Who / where / how you teach

OXE

WHO

Many robots / many labs soup

DROID

WHERE

Same Franka · many real scenes

ALOHA

HOW (lab)

Leader-follower joint teleop

UMI

HOW (wild)

Human handheld · no robot logistics

BEHAVIOR

PROGRESS

Long-horizon household — steal UMI interface + modern heads

Scope note

Research + apps only — no Discord/Drive checklist

This deck stays on BEHAVIOR priors, complementary poles, CLEAR citations, follow-up science, and new applications.

No Discord channels, Drive folders, or ops checklists — those are out of scope for Visionary (per brief).

Numbers tattoo — pin these

Every deck carries the same spine numbers

100% / 90%

cup UR5 / Franka

87.5%

toss w/ latency

57.5%

toss no latency

71.7%

wild 43/60 OOD

0%

narrow → wild OOD

100% vs 25%

rel vs abs action

55%

no fisheye (69°)

70% / 30%

cloth ± inter-grip

70% / 0%

dish ViT / ResNet

1400 / 30

wild demos / locs

Ta=6

@10Hz (20Hz toss)

73 + 298

BOM · 780g

Exam bite — Visionary

UMI turns in-the-wild from robot logistics into human logistics. Complementary to DROID & ALOHA. $\pi_{0.5}$ /GR00T cite it. Steal the interface (relative+latency+fisheye); race modern FM/VLA heads — don't ship 2024 ResNet DP.

Visionary — close

Takeaways + exam bite

1. BEHAVIOR prior: cheap handheld scene diversity without Franka fleets.
2. Poles: UMI human-wild || DROID robot-wild || ALOHA lab teleop.
3. Follow-ups: embodiment-aware, FM swap, RTC, mobile, distributed MP4.
4. One-liner: robot logistics → human logistics; $\pi_{0.5}$ /GR00T still cite UMI.

EXAM BITE

Human logistics; complementary poles; cite CLEAR; steal interface not 2024 DP head.